

UNIT:01

Introduction to Data Wrangling

Data wrangling is a crucial initial step in any data analysis, visualization, or machine learning project. It involves transforming raw data into a format that is more suitable for analysis. This process includes a series of steps, such as cleaning, transforming, and enriching the data to ensure that it is structured and reliable.

Definition of Data Wrangling

Data wrangling, also known as data munging, is the process of converting raw data into a structured and clean format that is ready for analysis. It encompasses various activities aimed at handling inconsistencies, missing data, and ensuring that the data meets the necessary quality standards for reliable analysis.

Importance of Data Wrangling

The quality of your analysis is highly dependent on the quality of your data. Even sophisticated algorithms can yield poor results if the input data is flawed. Here are a few reasons why data wrangling is essential:

1. Ensures Data Quality:

- High-quality data is accurate, consistent, and complete. Data wrangling helps identify and rectify issues such as errors, inconsistencies, and outliers.

2. Enhances Analysis:

- Clean and structured data allows analysts and data scientists to focus on deriving insights rather than spending time dealing with data issues.

3. Reduces Complexity:

- By transforming data into a more usable form, data wrangling simplifies complex datasets, making them easier to understand and analyze.

4. Improves Model Performance:

- Properly wrangled data improves the performance of machine learning models, as these models rely on clean, normalized, and standardized data.

5. Facilitates Reproducibility:

- Data wrangling often involves documenting each transformation step, which makes the process reproducible and transparent for other analysts or stakeholders.

Common Data Wrangling Tasks

Data wrangling involves a series of tasks, each aimed at handling specific types of issues found in raw data:

1. Data Cleaning:

- Removing Duplicates: Eliminating repeated entries to avoid skewed analysis.
- Handling Missing Values: Addressing gaps in data using techniques like imputation or removal.
- Correcting Errors: Fixing inaccuracies in data such as incorrect spellings, erroneous values, etc.

2. Data Transformation:

- Normalizing and Standardizing: Adjusting data values to a common scale.
- Encoding Categorical Variables: Converting categories into numerical values for machine learning models.
- Aggregating Data: Summarizing data to capture trends, such as computing monthly sales from daily records.

3. Data Enrichment:

- Adding Contextual Information: Incorporating additional data, such as weather information for sales data or demographic information for customer records.
- Creating New Features: Generating new variables from existing data, such as calculating customer lifetime value or creating time-based features.

4. Data Integration:

- Merging Datasets: Combining data from different sources to create a unified dataset.
- Resolving Conflicts: Addressing discrepancies between data from different sources.

5. Data Validation:

- Consistency Checks: Ensuring that data adheres to defined rules, such as dates being in a valid range or numerical fields not containing text.
- Cross-Validation: Comparing data against external sources or historical trends to verify accuracy.

Tools and Techniques for Data Wrangling

Data wrangling can be performed using various tools and techniques, depending on the complexity of the data and the specific requirements of the analysis:

1. Programming Languages:

- Python: Widely used for data wrangling with libraries like Pandas, NumPy, and Dask.
- R: Provides powerful packages like dplyr, tidyr, and data.table for data manipulation.

2. Visualization Tools:

- Tableau/Power BI: Useful for visual data cleaning and validation.
- Matplotlib/Seaborn: Used in Python for visual exploration of data distributions and anomalies.

3. Data Profiling Tools:

- Pandas Profiling: Automatically generates comprehensive data profiling reports in Python.
- DataPrep: A Python library for easy EDA and data cleaning.
- Great Expectations: A tool for validating, documenting, and profiling your data.

4. Spreadsheets:

- Excel/Google Sheets: Often used for manual data cleaning and exploration, especially for smaller datasets.

Real-World Examples of Data Wrangling

1. Business Analytics:

- A retail company collects data from multiple sources, including sales records, customer interactions, and online behavior. Data wrangling is used to clean and merge these datasets, enabling comprehensive analysis of customer behavior and sales trends.

2. Healthcare:

- Medical researchers use data wrangling to clean and integrate patient records from various hospitals. This allows for consistent analysis of treatment outcomes across different demographic groups.

3. Finance:

- Financial analysts wrangle data from stock markets, economic indicators, and company reports to build predictive models for investment strategies.

4. Marketing:

- Marketing teams clean and merge data from CRM systems, social media platforms, and email campaigns to better understand customer engagement and optimize marketing efforts.
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Challenges in Data Wrangling

Data wrangling can be a complex and time-consuming process, often requiring a deep understanding of the data and domain-specific knowledge. Common challenges include:

1. Handling Large Datasets:

- Working with large volumes of data can be resource-intensive and require optimization techniques to ensure efficient processing.

2. Dealing with Unstructured Data:

- Unstructured data like text, images, or social media feeds need specialized techniques for cleaning and transformation.

3. Data Privacy and Security:

- Ensuring that sensitive information is handled appropriately, especially when dealing with personal or financial data.

4. Automating the Wrangling Process:

- Creating automated, reusable pipelines for data wrangling can be challenging but is essential for scalability and efficiency.

5. Changing Data Sources:

- Data sources can change over time, requiring ongoing adjustments to data wrangling processes to maintain data quality and consistency.

Need for Data Cleanup

Data cleanup, also known as data cleansing or data cleaning, is an essential step in the data wrangling process. It involves identifying and correcting errors, inconsistencies, and inaccuracies in the data to ensure that it is of high quality and suitable for analysis. High-quality data is crucial for making reliable and accurate decisions, as poor data can lead to misleading conclusions and ineffective models.

Common Issues in Raw Data

Raw data often comes with a variety of issues that need to be addressed before it can be used effectively. These issues can arise from different sources such as data entry errors, system errors, or inconsistencies between different data sources. Here are some common problems found in raw data:

1. Incomplete Data:

- Definition: Missing values in a dataset, which can occur when no data value is stored for a variable in an observation.
- Causes:
 - Data entry errors or omissions.
 - Non-response in surveys.
 - Data loss during transmission or storage.
- Impact: Incomplete data can lead to biased analysis and affect the performance of machine learning models.
- Example: In a dataset of customer information, some entries may lack email addresses or phone numbers.

2. Inconsistent Data:

- Definition: Data that lacks uniformity in format, type, or value, making it difficult to compare or aggregate.
- Causes:
 - Different data entry standards (e.g., date formats like MM/DD/YYYY vs. DD/MM/YYYY).
 - Variations in categorical data (e.g., "Male" vs. "M" vs. "male").
 - Data collected from multiple sources with different formats.
- Impact: Inconsistent data can lead to errors in data aggregation, filtering, and analysis.

- Example: In a dataset of product sales, the product category might be recorded as "electronics" in one entry and "Electronics" in another.

3. Duplicate Data:

- Definition: Repeated entries for the same observation, which can result from data entry errors or merging datasets without proper checks.
- Causes:
 - Manual data entry or batch processing errors.
 - Multiple data sources recording the same information.
- Impact: Duplicates can skew statistical analysis, leading to incorrect insights.
- Example: A customer's information is recorded twice in a CRM system, resulting in inflated customer count and sales figures.

4. Erroneous Data:

- Definition: Data that contains incorrect or implausible values.
- Causes:
 - Typographical errors during data entry.
 - Misconfigurations in data collection systems.
- Impact: Erroneous data can distort analysis and lead to incorrect conclusions.
- Example: An age field containing negative values or a transaction date that is set in the future.

5. Outliers:

- Definition: Extreme values that deviate significantly from other observations in the dataset.
- Causes:
 - Data entry mistakes.
 - Genuine but rare events (e.g., a sudden surge in sales due to a marketing campaign).
- Impact: Outliers can skew statistical measures such as mean and standard deviation and affect the performance of machine learning models.

- Example: A transaction value of \$1,000,000 in a dataset where the average transaction value is \$50.

Impacts of Unclean Data

Unclean data can have several negative consequences, impacting both analysis and decision-making processes:

1. Inaccurate Analysis:

- Analysis based on flawed data can lead to incorrect conclusions and poor decision-making. For example, missing or incorrect data in sales records can result in a miscalculation of total sales and revenue projections.

2. Increased Processing Time:

- Dealing with unclean data requires additional preprocessing time and effort, delaying analysis and reporting. This is particularly problematic in industries where timely insights are critical, such as finance or healthcare.

3. Reduced Model Performance:

- Machine learning models trained on unclean data often perform poorly. Models may overfit to noisy data, miss important patterns, or generate biased predictions, leading to ineffective outcomes.

4. Misleading Visualizations:

- Data visualizations based on unclean data can be misleading, causing stakeholders to misinterpret trends or relationships in the data.

5. Inconsistent Reports:

- When data quality issues are not addressed, different reports based on the same data can yield conflicting results, undermining trust in the data and the insights derived from it.

Steps to Identify Data Quality Issues

Identifying data quality issues is a critical part of the data wrangling process. It involves systematically examining the data to uncover and understand potential problems.

1. Data Profiling:

- Definition: The process of reviewing the data to understand its structure, content, and quality. This includes analyzing metadata (e.g.,

column names, data types) and generating descriptive statistics (e.g., mean, median, mode, range).

- Techniques:
 - Summary Statistics: Calculating metrics like mean, median, standard deviation, and count of missing values for each column.
 - Data Distribution Analysis: Visualizing the distribution of values using histograms or box plots to identify skewness, outliers, and anomalies.
 - Relationship Analysis: Assessing correlations and dependencies between variables to identify inconsistencies or unexpected patterns.

2. Data Validation:

- Definition: The process of checking data for accuracy and consistency against predefined rules or standards.
- Techniques:
 - Domain Checks: Ensuring that values fall within an acceptable range (e.g., age between 0 and 120).
 - Format Checks: Verifying that data adheres to a specific format (e.g., email addresses contain an "@" symbol).
 - Business Rule Validation: Ensuring that data complies with business rules (e.g., order date must be earlier than delivery date).

3. Exploratory Data Analysis (EDA):

- Definition: An approach for analyzing data sets to summarize their main characteristics, often using visual methods.
- Techniques:
 - Visualization: Use of graphs and plots (e.g., scatter plots, bar charts) to explore data distributions and identify anomalies.
 - Cross-Tabulation: Analyzing the relationship between two or more categorical variables to identify inconsistencies or unexpected patterns.

- Descriptive Statistics: Calculating statistics like mean, median, variance, and skewness to understand the data's central tendency and dispersion.

4. Automated Data Quality Checks:

- Definition: Using software tools to automatically identify and report data quality issues.
- Tools:
 - Pandas Profiling: Automatically generates a detailed report of data quality issues in a pandas DataFrame.
 - Great Expectations: A Python library for creating, testing, and documenting data expectations.
 - Data Prep: A Python library for EDA and data cleaning.

Examples of Data Cleanup in Different Domains

1. Healthcare:

- Scenario: Patient records from different hospitals are merged for a study. The dataset has missing values for patient age and inconsistent formats for test results.
- Cleanup Process:
 - Fill missing age values using imputation (e.g., median age).
 - Standardize test result units (e.g., converting mg/dL to mmol/L).
 - Remove duplicate patient records.

2. Finance:

- Scenario: A financial analyst is working with a dataset of daily stock prices. The dataset has missing values on weekends and erroneous data entries with negative stock prices.
- Cleanup Process:
 - Remove weekends and holidays to focus on trading days.
 - Correct negative prices using external validation or remove affected entries.

3. Retail:

- Scenario: A retail company wants to analyze customer purchase behavior. The dataset contains duplicated entries and inconsistent product categories.
- Cleanup Process:
 - Remove duplicate customer entries based on unique identifiers.
 - Standardize product categories (e.g., "electronics" vs. "Electronics").
 - Impute missing purchase amounts with the average purchase value.

4. Marketing:

- Scenario: A marketing team is analyzing email campaign data. The dataset has missing email addresses and inconsistent response rates.
- Cleanup Process:
 - Remove records with missing email addresses.
 - Normalize response rate data by standardizing the format and converting to a consistent scale.

Best Practices for Data Cleanup

1. Document the Cleanup Process:

- Keep detailed records of the steps taken to clean the data, including justifications for handling specific issues. This ensures reproducibility and transparency.

2. Use Automated Tools Where Possible:

- Leverage tools and libraries to automate repetitive tasks, such as missing value imputation or data validation. This can save time and reduce errors.

3. Iterate and Validate:

- Data cleaning is often an iterative process. Perform initial cleanup, validate the results, and refine as needed.

4. Involve Domain Experts:

- Collaborate with domain experts to understand the context of the data and make informed decisions during the cleanup process.

5. Create Reusable Cleaning Pipelines:

- Develop reusable scripts or pipelines for data cleaning, especially when dealing with frequently updated datasets. This ensures consistency and efficiency.

Data Cleanup Basics

Data cleanup is a foundational component of the data wrangling process. It ensures that the data used for analysis is accurate, consistent, and reliable.

Removing Missing Values

Missing values can severely impact data analysis and decision-making. It is crucial to identify and address them appropriately.

- **Identification:**
 - Use functions like `isnull()` in Python or `is.na()` in R to detect missing values in datasets.
 - Generate summary statistics to determine the proportion of missing values per column.
- **Strategies for Handling Missing Values:**
 - **Deletion:**
 - **Row Deletion:** Remove entire rows with missing values if they constitute a small portion of the dataset.
 - **Column Deletion:** Remove columns with a high percentage of missing values, particularly if they are not essential to the analysis.
 - **Imputation:**
 - **Mean/Median/Mode Imputation:** Fill in missing values using the mean, median, or mode of the non-missing values in the column. This is a simple yet effective method, particularly for numerical data.
 - **Predictive Imputation:** Use machine learning algorithms (e.g., regression models) to predict and fill in missing values based on other features.

- **Forward/Backward Fill:**
 - This technique is commonly used in time-series data, where the previous or subsequent value is used to fill in gaps (e.g., `fillna(method='ffill')` in Python).

Handling Outliers

Outliers are data points that deviate significantly from other observations, and they can distort analysis and statistical measures.

- **Identification:**
 - **Visual Methods:** Use box plots or scatter plots to visually detect outliers.
 - **Statistical Methods:** Calculate Z-scores or apply the Interquartile Range (IQR) method:
 - Z-score method: A data point is considered an outlier if its Z-score is greater than 3 or less than -3.
 - IQR method: Identify outliers as values lying beyond $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$.
- **Strategies for Handling Outliers:**
 - **Capping:** Limit extreme values to a defined maximum or minimum threshold to reduce their impact on analysis.
 - **Transformation:** Apply transformations (e.g., log, square root) to reduce the influence of outliers.
 - **Removal:** Exclude outliers if they are determined to be errors or if they do not provide relevant information for analysis.

Dealing with Duplicates

Duplicate entries can lead to inflated counts and distorted analysis, making it vital to identify and remove them.

- **Identification:**
 - Use functions like `duplicated()` in Python or `unique()` in R to find duplicate records in a dataset.
 - Analyze key columns to determine duplicates, especially in unique identifiers like IDs or emails.

- **Removal:**
 - **Exact Duplicate Removal:** Use methods to drop exact duplicates while retaining one instance of the record.
 - **Near-Duplicate Handling:** Define criteria for near-duplicates (e.g., slight variations in text) and consolidate them into a single entry.

Correcting Data Types

Ensuring that data types are correctly assigned is crucial for accurate analysis and modeling.

- **Identification:**
 - Use functions to check the data types of each column (e.g., dtypes in Pandas).
- **Common Issues:**
 - **Numeric vs. Categorical:** Numerical data may be stored as strings, leading to errors in calculations.
 - **Date Formats:** Date values may be misformatted, requiring conversion to a proper date type.
- **Conversions:**
 - Use functions like `astype()` in Python or `as.numeric()` in R to convert data types as needed.
 - Apply functions like `pd.to_datetime()` in Pandas for converting strings to datetime objects.

1.3.5 Data Standardization and Normalization

Standardization and normalization are techniques used to scale and prepare data for analysis.

- **Normalization:**
 - **Definition:** Adjusting values to a common scale, typically [0, 1].
 - **Min-Max Scaling:**
$$X_{\text{norm}} = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}}$$
 - **Application:** Useful when you want to compare data points from different ranges (e.g., income vs. age).

- **Standardization:**

- **Definition:** Transforming data to have a mean of 0 and a standard deviation of 1.
- **Z-Score Standardization:** $Z = \frac{X - \mu}{\sigma}$
- **Application:** Effective for normally distributed data, allowing comparisons across different scales.

Best Practices for Data Cleanup

1. **Document the Cleaning Process:** Keep detailed records of cleaning steps to ensure reproducibility and understanding of decisions made.
2. **Iterative Approach:** Data cleaning is often an iterative process. Perform initial cleaning, validate, and refine.
3. **Collaboration with Domain Experts:** Engage with subject matter experts to ensure that data cleaning aligns with business requirements and contextual understanding.
4. **Use Automated Tools:** Leverage libraries and tools to automate repetitive tasks, reducing manual errors and saving time.
5. **Visualize Data After Cleanup:** Use visualizations to validate the effectiveness of data cleaning efforts and ensure that anomalies have been addressed.

Practical Examples

1. **Example of Removing Missing Values:**

- Dataset: Customer information with missing phone numbers.
- Process: Use mean imputation to fill missing phone number values based on customer demographics.

2. **Example of Handling Outliers:**

- Dataset: Sales data with a few extremely high transaction values.
- Process: Identify outliers using box plots, cap extreme values at the 95th percentile.

3. **Example of Removing Duplicates:**

- Dataset: Email campaign results with duplicate entries for some customers.

- Process: Identify duplicates based on email addresses and retain unique entries.

4. Example of Correcting Data Types:

- Dataset: Product catalog with price stored as strings.
- Process: Convert the price column to numeric data types for proper analysis.

5. Example of Normalizing Data:

- Dataset: User activity data with varying scales for different metrics.
- Process: Apply Min-Max scaling to normalize user engagement scores.

Data cleanup is a critical step in the data wrangling process, directly impacting the quality and reliability of subsequent analysis. By systematically addressing missing values, outliers, duplicates, and data types, analysts can ensure that their datasets are primed for accurate insights. Utilizing best practices and appropriate tools enhances the efficiency and effectiveness of the data cleanup process, ultimately leading to better decision-making.

Normalizing and Standardizing Data

Normalization and standardization are vital techniques in data preprocessing that prepare datasets for analysis and machine learning. These processes help ensure that the data is on a consistent scale, which is crucial for many algorithms that are sensitive to the scale of input features.

Definitions

- **Normalization:**

- The process of scaling individual data points to a common range, usually $[0, 1]$.
- It is particularly useful for datasets where features have different units or vastly different scales.

- **Standardization:**

- The process of transforming data to have a mean of 0 and a standard deviation of 1.
- This technique is effective when the data follows a Gaussian (normal) distribution.

Why Normalize or Standardize Data?

1. Improved Model Performance:

- Many machine learning algorithms, such as k-means clustering and gradient descent-based methods, are sensitive to the scale of the input data. Normalizing or standardizing helps improve their performance and convergence speed.

2. Ensuring Equal Weight:

- When features are on different scales, some may disproportionately influence the model's predictions. Normalization and standardization help to mitigate this issue.

3. Facilitating Comparisons:

- Normalized or standardized data allows for more straightforward comparisons between different features or datasets.

4. Handling Outliers:

- Standardization is less affected by outliers than normalization, making it a preferred choice when outliers are present.

Methods of Normalization

1. Min-Max Scaling:

- This technique rescales the feature to a fixed range, typically [0, 1].
- Formula:**
$$X_{\text{norm}} = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}}$$
- Advantages:**
 - Preserves the relationships between data points.
 - Useful for algorithms requiring bounded input (e.g., neural networks).
- Disadvantages:**
 - Sensitive to outliers, as they can skew the min and max values.

2. Robust Scaling:

- This technique scales features using statistics that are robust to outliers, such as the median and interquartile range (IQR).

- **Formula:** $X_{\text{robust}} = \frac{X - \text{median}}{\text{IQR}}$
- **Advantages:**
 - Less affected by outliers compared to min-max scaling.
- **Disadvantages:**
 - Does not necessarily bound the data.

Methods of Standardization

1. Z-score Standardization:

- This method transforms the data such that it has a mean of 0 and a standard deviation of 1.
- **Formula:** $Z = \frac{X - \mu}{\sigma}$ Where μ is the mean and σ is the standard deviation.
- **Advantages:**
 - Useful for algorithms that assume normally distributed data (e.g., linear regression).
 - Reduces the effect of outliers.
- **Disadvantages:**
 - Not bounded; standardized data can take on any value.

2. Robust Z-score:

- A variant that uses the median and median absolute deviation instead of mean and standard deviation to reduce the influence of outliers.
- **Formula:** $Z_{\text{robust}} = \frac{X - \text{median}}{\text{MAD}}$ Where MAD is the median absolute deviation.
- **Advantages:**
 - More robust to outliers compared to standard Z-score.

When to Use Normalization vs. Standardization

- **Use Normalization:**

- When the dataset has features on different scales and requires a bounded range.
- In algorithms that rely on distance metrics (e.g., k-nearest neighbors) or neural networks.
- When the data contains outliers that you wish to limit in influence.
- **Use Standardization:**
 - When the data follows a Gaussian distribution or you want to achieve a zero mean and unit variance.
 - In algorithms that assume normality (e.g., linear regression, logistic regression).
 - When outliers are present, and you want to mitigate their effect while retaining them in the dataset.

Implementation Examples:

1.Normalization in Python (Pandas):

```
import pandas as pd  
  
# Sample data  
data = {'feature1': [1, 2, 3, 4, 5], 'feature2': [10, 20, 30, 40, 50]}  
df = pd.DataFrame(data)  
  
# Min-Max Normalization  
df_normalized = (df - df.min()) / (df.max() - df.min())
```

2. Standardization in Python (Pandas):

```
import pandas as pd  
  
# Sample data
```

```
data = {'feature1': [1, 2, 3, 4, 5], 'feature2': [10, 20, 30, 40, 50]}  
df = pd.DataFrame(data)  
  
# Z-score Standardization  
df_standardized = (df - df.mean()) / df.std()
```

3.Using Scikit-learn for Scaling:

```
from sklearn.preprocessing import MinMaxScaler, StandardScaler
```

```
# Sample data
```

```
data = [[1, 10], [2, 20], [3, 30], [4, 40], [5, 50]]
```

```
# Normalization
```

```
min_max_scaler = MinMaxScaler()
```

```
normalized_data = min_max_scaler.fit_transform(data)
```

```
# Standardization
```

```
standard_scaler = StandardScaler()
```

```
standardized_data = standard_scaler.fit_transform(data)
```

Best Practices for Normalization and Standardization

1. **Know Your Data:** Understand the distribution of your data and the presence of outliers before deciding on normalization or standardization.

2. **Apply Consistently:** Apply the same normalization or standardization method across all training and test datasets to avoid data leakage and inconsistencies.
3. **Document Your Process:** Maintain clear documentation of the scaling method used and the rationale behind choosing a particular technique.
4. **Visualize Results:** Use histograms, box plots, and scatter plots to visualize the effects of normalization or standardization on the data.
5. **Test Different Methods:** Experiment with both normalization and standardization in modeling to see which yields better performance for your specific use case.

Normalizing and standardizing data are critical steps in the data wrangling process that enhance the effectiveness of data analysis and machine learning models. By ensuring that data is on a consistent scale, analysts can improve model performance and ensure accurate comparisons across features. Understanding when and how to apply these techniques is essential for effective data preparation.

Data Transformation Techniques

Data transformation involves converting data from one format or structure to another. This is a vital process in data wrangling, enhancing the quality and usability of data for analysis, visualization, and machine learning. This module discusses various data transformation techniques, their significance, and applications.

Importance of Data Transformation

Data transformation plays a critical role in the data wrangling process by:

1. **Improving Data Quality:** It helps rectify inconsistencies, errors, and discrepancies within datasets, leading to more reliable analysis.
2. **Facilitating Analysis:** Transforming data into a suitable format makes it easier to analyze and interpret, ensuring that the dataset aligns with analytical techniques.
3. **Enabling Data Integration:** Data from multiple sources often have different formats. Transformation standardizes these formats, allowing for seamless integration.
4. **Enhancing Model Performance:** Machine learning models often perform better with well-structured data, as certain algorithms require features to be on similar scales or distributions.

Common Data Transformation Techniques:

1. Data Aggregation

- **Definition:** Data aggregation involves summarizing a dataset by grouping data points based on one or more keys and applying aggregate functions (such as sum, mean, count, etc.).
- **Purpose:**
 - Reduces the size of large datasets, making them easier to analyze.
 - Retains important summary statistics that provide insights into trends and patterns.
- **Applications:**
 - Monthly or yearly sales reports from daily sales data.
 - Summarizing survey responses by demographic categories.

2. Data Encoding

- **Definition:** Data encoding converts categorical variables into a numerical format, allowing algorithms to interpret categorical data effectively.
- **Types:**
 - **Label Encoding:** Assigns a unique integer to each category.
 - **Purpose:** Simple and efficient for ordinal categories (where order matters).
 - **Example:** "Low" = 1, "Medium" = 2, "High" = 3.
 - **One-Hot Encoding:** Creates binary columns for each category, indicating the presence (1) or absence (0) of each category.
 - **Purpose:** Useful for nominal categories where no inherent order exists.
 - **Example:** For the color feature (red, green, blue), it generates three binary columns.

3. Data Scaling

- **Definition:** Data scaling adjusts the range of feature values to ensure that they contribute equally to distance calculations in algorithms, particularly for those sensitive to feature magnitudes.
- **Techniques:**
 - **Min-Max Scaling:** Rescales features to a fixed range, typically [0, 1].

- Formula:
$$X_{\text{scaled}} = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}}$$
- Use Case: Useful for algorithms that require bounded input, such as neural networks.
- Z-score Standardization (Standardization): Transforms data to have a mean of 0 and a standard deviation of 1.
 - Formula:
$$Z = \frac{X - \mu}{\sigma}$$
 Where μ is the mean and σ is the standard deviation.
 - Use Case: Suitable for algorithms that assume normally distributed data, such as linear regression.

4. Data Discretization

- Definition: Data discretization involves converting continuous variables into discrete categories or bins.
- Purpose:
 - Simplifies the model by reducing the number of unique values.
 - Enhances interpretability by converting continuous ranges into categorical groups.
- Applications:
 - Grouping ages into categories like "child," "teen," "adult," and "senior."
 - Creating income brackets for analysis.

5. Feature Engineering

- Definition: Feature engineering is the process of creating new features from existing ones to improve model performance.
- Techniques:
 - Polynomial Features: Involves creating new features by raising existing features to a power or creating interaction terms (e.g., $X_1 \times X_2$).
 - Date/Time Features: Extracting components such as year, month, day, day of the week, etc., from datetime variables.
 - Text Features: Converting text data into numerical representations, such as term frequency or using embeddings.

6. Mathematical Transformations

- Definition: Mathematical transformations involve applying functions to modify the distribution of data.
- Common Functions:
 - Log Transformation: Reduces the skewness of right-skewed data, making it more normally distributed.
 - Use Case: Effective for datasets with exponential growth patterns.
 - Square Root Transformation: Can also help reduce skewness, particularly for count data.
 - Box-Cox Transformation: A family of power transformations that can stabilize variance and make data more normally distributed.

Data transformation techniques are fundamental to data wrangling, enabling analysts to prepare datasets for effective analysis and modeling. Understanding when and how to apply these techniques enhances data quality and improves the performance of analytical models. Each technique serves specific purposes and is chosen based on the characteristics of the dataset and the requirements of the analysis.